Evaluating Trigonometric Functions

Name _____

Algebra 2 Lesson 7-2 (fast track) Date _____ Period _____

Learning goal: I can use special triangles, reference angles, and the Pythagorean identity to find the trigonometric values of an angle in any quadrant.

Language goal: I can explain how I found a trigonometric value using the words reference angle, quadrant, terminal point, and reciprocal.

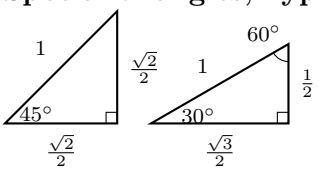
On the quiz: (1) Sine and cosine of a special angle such as $2\pi/3$ (Example 4). (2) Sine from cosine and the quadrant (Example 6). (4) Bonus: cosecant, cotangent and secant of 225° (Example 4A, the reciprocal box, the Practice). (3) and (5): full Savvas quiz only.

On the Topic 7 test: Question 1 asks for $\sec\theta$ from $\sin\theta$, Question 3 for $\sin\theta$ from $\cot\theta$, and Question 16 for $\cos\theta$ from $\sin\theta$ and the quadrant (Example 6). Question 4 asks for the sine and cosine of $4\pi/3$ (Example 4).

Example 4: Evaluate Trigonometric Functions with Special Angles as Reference Angles

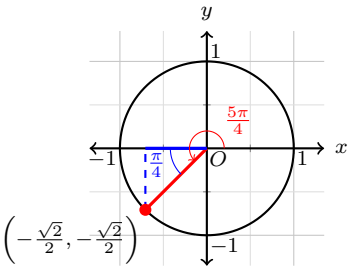
1. Find the **quadrant** of the terminal side.
2. Find the **reference angle**: the acute angle between the terminal side and the x-axis.
3. A special triangle (hypotenuse 1) gives the legs; the quadrant gives the signs. That is the **terminal point** $(x, y) = (\cos\theta, \sin\theta)$.
4. $\sin\theta = y$, $\cos\theta = x$, $\tan\theta = \frac{y}{x}$.

Special triangles, hypotenuse 1 (Example 3)



30°-60°-90°: short leg = half the hypotenuse; long leg = short leg $\times\sqrt{3}$.

A. What are the sine, cosine, and tangent of $\frac{5\pi}{4}$?

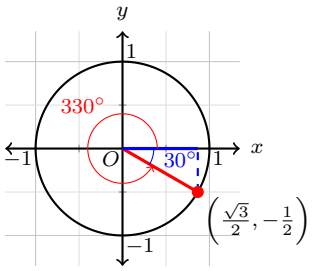


Quadrant III. Reference angle $\frac{5\pi}{4} - \pi = \frac{\pi}{4}$, or 45°.

$\sin\frac{5\pi}{4} = -\frac{\sqrt{2}}{2}$, $\cos\frac{5\pi}{4} = -\frac{\sqrt{2}}{2}$

$\tan\frac{5\pi}{4} = \frac{-\sqrt{2}/2}{-\sqrt{2}/2} = 1$

B. What are the sine, cosine, and tangent of 330°?

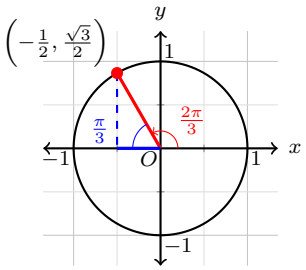


Quadrant IV. Reference angle $360° - 330° = 30°$, or $\frac{\pi}{6}$.

$\sin 330° = -\frac{1}{2}$, $\cos 330° = \frac{\sqrt{3}}{2}$

$\tan 330° = \frac{-1/2}{\sqrt{3}/2} = -\frac{1}{\sqrt{3}} = -\frac{\sqrt{3}}{3}$

C. What are the sine, cosine, and tangent of $\frac{2\pi}{3}$?



Quadrant II. Reference angle $\pi - \frac{2\pi}{3} = \frac{\pi}{3}$, or 60°.

$\sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$, $\cos\frac{2\pi}{3} = -\frac{1}{2}$

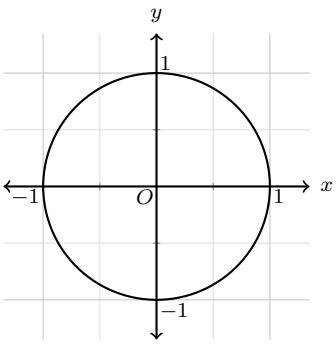
$\tan\frac{2\pi}{3} = \frac{\sqrt{3}/2}{-1/2} = -\sqrt{3}$

Say it: The terminal side of 330° is in Quadrant _____, so the reference angle is _____. The terminal point is _____, and $\sin 330°$ is negative because

Try It 4: Find the sine, cosine, and tangent of angles $\frac{4\pi}{3}$ and $-210°$.

Hint: Sketch each angle and its reference triangle on the unit circle, then fill in the steps. A negative angle turns clockwise.

a. $\theta = \frac{4\pi}{3}$



Quadrant: _____

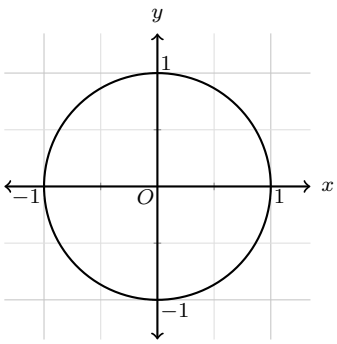
Reference angle: _____

Terminal point: (_____, _____)

$\sin\frac{4\pi}{3} =$ _____ $\cos\frac{4\pi}{3} =$ _____

$\tan\frac{4\pi}{3} =$ _____

b. $\theta = -210°$



Quadrant: _____

Reference angle: _____

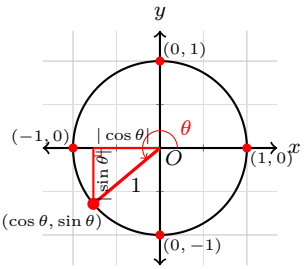
Terminal point: (_____, _____)

$\sin(-210°) =$ _____ $\cos(-210°) =$ _____

$\tan(-210°) =$ _____

Example 6: Use the Pythagorean Identity $\sin^2\theta + \cos^2\theta = 1$

A. Prove the Pythagorean Identity $\sin^2\theta + \cos^2\theta = 1$.



Off the axes, the reference triangle has legs $|x|$, $|y|$ and hypotenuse 1. Pythagorean Theorem: $|y|^2 + |x|^2 = 1$, so $y^2 + x^2 = 1$

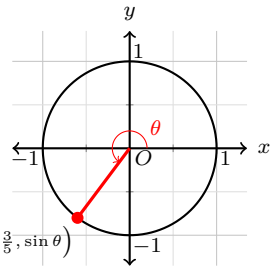
$(\sin\theta)^2 + (\cos\theta)^2 = 1$

$\sin^2\theta + \cos^2\theta = 1$

Substitute $y = \sin\theta$ and $x = \cos\theta$. On an axis, $x^2 + y^2 = 1$ directly.

$\sin^2\theta$ means $(\sin\theta)(\sin\theta)$, not $\sin(\theta^2)$.

B. What are $\sin\theta$ and $\tan\theta$ if $\cos\theta = -\frac{3}{5}$ and the terminal side of θ is in Quadrant III?



$\sin^2\theta + \cos^2\theta = 1$

$\sin^2\theta + \left(-\frac{3}{5}\right)^2 = 1$

$\sin^2\theta + \frac{9}{25} = 1$

$\sin^2\theta = \frac{16}{25}$

$\sin\theta = \pm\frac{4}{5}$

Quadrant III: the y-coordinate is negative, so $\sin\theta = -\frac{4}{5}$.

$$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{-\frac{4}{5}}{-\frac{3}{5}} = \frac{4}{3}$$

Reciprocals (from Example 5, which this packet skips): $\csc\theta = \frac{1}{\sin\theta} = \frac{1}{y}$ $\sec\theta = \frac{1}{\cos\theta} = \frac{1}{x}$ $\cot\theta = \frac{1}{\tan\theta} = \frac{x}{y}$. A **reciprocal** has the same sign as its function.

Say it: The terminal side of θ is in Quadrant III, so the _____-coordinate of the **terminal point** is negative and $\sin\theta =$ _____. The secant is the _____ of the cosine, so $\sec\theta =$ _____ because

Try It 6: Use the Pythagorean Identity

a. For θ in Quadrant IV, find $\tan\theta$ if $\sin\theta = -\frac{8}{17}$.

$\cos\theta =$ _____ $\tan\theta =$ _____

b. For θ in Quadrant II, find $\cos\theta$ if $\tan\theta = -\frac{12}{5}$.

$\cos\theta =$ _____

Hint: Put the value you know into $\sin^2\theta + \cos^2\theta = 1$, then let the quadrant choose the sign. In b, $\tan\theta = \frac{\sin\theta}{\cos\theta}$, so $\sin\theta = -\frac{12}{5}\cos\theta$.

In part a, why is $\cos\theta$ positive? Use the words *quadrant* and *terminal point*.

Practice: Reciprocals outside Quadrant I

a. Describe and correct the error a student made in evaluating the secant of a 135° angle.

> The coordinates of the terminal point on the unit circle are $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$.
>
> $$\sec 135° = \frac{-\frac{\sqrt{2}}{2}}{1} = -\frac{\sqrt{2}}{2}$$ ×

b. Find the secant, cosecant, and cotangent of a 135° angle.

Hint: Quadrant? Reference angle? Terminal point? Then take reciprocals.

$\sec 135° =$ _____ $\csc 135° =$ _____ $\cot 135° =$ _____

c. If $\sin\theta = \frac{5}{13}$ and $\tan\theta < 0$, what is the value of $\sec\theta$? Explain.

Hint: Which quadrant has $\sin\theta > 0$ and $\tan\theta < 0$? Find $\cos\theta$ with the identity, then take its reciprocal.

$\cos\theta =$ _____ $\sec\theta =$ _____

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first. **SmartScore 70 or above on each starred (★) skill replaces your quiz score.**

Code	Skill	Feeds
6H8 ★	Evaluate trigonometric ratios using reference angles: sin, cos, tan	Quiz 1 (Example 4); Topic test 4
XJJ ★	Find trigonometric ratios using a Pythagorean or reciprocal identity	Quiz 2 (Example 6); Topic test 1, 3, 16
PAE ★	Evaluate trigonometric ratios using reference angles: sec, csc, cot	Quiz 4 bonus (Example 4A and the reciprocal box)
NUF	Special right triangles	Warm-up: the 45° and 30°-60° legs
BRP	Reference angles	Warm-up: step 2 of Example 4

Exit question: Practice 19 (Error Analysis)

If the coordinates of the terminal point of an angle θ on the unit circle are $\left(-\frac{3}{5}, \frac{4}{5}\right)$, describe and correct the error a student made in finding $\tan\theta$.

$$\tan\theta = \frac{-\frac{3}{5}}{\frac{4}{5}} = -\frac{3}{4} \quad \times$$

The terminal point is $(x, y) = \left(_____,\ _____\right)$, in Quadrant _____. The student divided _____ by _____, which gives the _____ of the tangent (the cotangent). The correct value is $\tan\theta = \frac{y}{x} =$ _____.

Before the quiz, finish these sentences.

The terminal point of θ on the unit circle is $\left(_____,\ _____\right)$.

The reference angle is the acute angle between the terminal side and the _____.

The _____ of the terminal side tells me the signs of $\sin\theta$ and $\cos\theta$.

$\sin^2\theta + \cos^2\theta =$ _____ because the hypotenuse of the reference triangle is _____.

$\csc\theta$, $\sec\theta$ and $\cot\theta$ are the _____ of $\sin\theta$, $\cos\theta$ and $\tan\theta$.

Spanish: *ángulo de referencia; secante, cosecante, cotangente; funciones trigonométricas recíprocas.*